PhD Math Camp 2026 – Syllabus

Department of Economics, Columbia University in the city of New York

Aug 10 – Aug 28, 2026

Course Information

- **Instructor:** Hao Jiang (haotian.jiang@columbia.edu).
- **Teaching Assistant:** Joe Jourden (jj3536@columbia.edu)
- **Website:** `https://sites.google.com/view/columbia-econ-math-camp-2026/home`.
- **Lecture:** 9:00 am–12:00 pm (with two breaks in between), Monday through Friday.
- **Recitation and Office Hour:** The instructor and the TA each hold weekly office hours, and the TA also holds weekly recitations. See the schedule below.
- **Location:** The first week is fully remote on Zoom. In the second and third week, all classes, office hours and recitations will be in-person and held in MLK 616. MLK classrooms are located inside Riverside Church building, on Claremont Ave between West 120th and 122nd street.

Introduction

This course provides a graduate-level introduction to the mathematical tools that appear regularly in the first-year PhD sequence. The course is rigorous in the sense that we will aim for clear definitions and careful statements. At the same time, it is not designed to be impossibly difficult, or to turn you into an expert in every topic within three weeks. The goal is to build familiarity, confidence, and a solid working foundation, so that when these tools arise later in coursework, they will feel recognizable rather than new.

The pace will be fairly quick, as we will cover a broad range of material in a short period. Still, you are not expected to master every detail immediately. What matters most is that you come away with a clear knowledge of the core results, a good sense of the main ideas, an understanding of the standard techniques, and a set of notes and references that you can return to when needed.

Target Audience and Prerequisites

This course is open to incoming first-year PhD students in Economics, Sustainable Development, Finance, Accounting and Quantitative Marketing at Columbia University. Students from other programs at Columbia University who wish to attend the course must obtain written approval from the department.

Prior knowledge of calculus, linear algebra and probability theory at the undergraduate level is expected. Please contact the instructor immediately if you feel that you do not meet this requirement, and we will find a solution.

Mode of Teaching and Learning

The first year of the PhD is, in general, a period of substantive learning and growth. While lectures are extremely important, a substantial part of learning happens outside the classroom through finding and consuming the right resources, working through problems, reviewing class materials, coming to class ready to engage actively, and above all, working together with your peers.

Our math camp follows this general model. Given the time constraints of the course, part of the learning will necessarily take place outside class. Video lectures (link below) recorded by Professor Mark Walker at the University of Arizona will be assigned in advance for each topic. Students are strongly encouraged to watch the relevant lecture recordings before class and to attend the in-person lectures (or online lectures during the first week) prepared with questions. This will lead to much better learning outcomes in this course and will also help build the habits that are essential for success in the first year. The link to the video lectures can be found here.

References

No single textbook covers all of the topics in this course, and the main course materials will be the self-contained lecture notes posted on the Math Camp website. Still, the following books cover most of the subjects treated in the course.

General Mathematics for Economists

- Simon, Carl P., and Lawrence Blume, *Mathematics for Economists* (1994).
- Sydsæter, Knut, Peter Hammond, Atle Seierstad, and Arne Strøm, *Further Mathematics for Economic Analysis* (2008).
- de la Fuente, Angel, *Mathematical Methods and Models for Economists* (2000).

Real analysis

- Corbae, Dean, Maxwell B. Stinchcombe, and Juraj Zápal, *An Introduction to Mathematical Analysis for Economic Theory and Econometrics* (2009).
- Ok, Efe A., *Real Analysis with Economic Applications* (2007).
- Rudin, Walter, *Principles of Mathematical Analysis* (1976).

Linear algebra

- Strang, Gilbert, *Introduction to Linear Algebra* (2016).
- Axler, Sheldon, *Linear Algebra Done Right* (2024).
- Treil, Sergei, *Linear Algebra Done Wrong* (2014).

Optimization

- Sundaram, Rangarajan K., *A First Course in Optimization Theory* (1996).
- Dixit, Avinash K., *Optimization in Economic Theory*, 2nd ed. (1990).

Probability

- Ross, Sheldon, *A First Course in Probability*, 10th ed. (2023).
- Blitzstein, Joseph K., and Jessica Hwang, *Introduction to Probability*, 2nd ed. (2019).

Course Evaluation

There is a graded closed-book exam at the end of the course, held in person on Friday, August 28, from 9:00 am to 12:00 pm. The exam grade is going to be 100% of the final grade. The exam is intended to assess whether you have developed a solid command of the core ideas and techniques covered in class. It is designed to be fair and serious, but fully in line with the material emphasized in the lectures, notes, and exercises.

A large set of exercises well be posted on the course website. During the course, two problem sets based on those exercises, and potentially additional practice problems, will be assigned. The assignments and their due dates are listed on the website under the *Problem Sets* tab. You may collaborate with other students to discuss and solve the problems, but each student must write up their own solution. Assignments should be submitted to the TA by email.

The problem sets are intended primarily as practice. They are meant to help you engage seriously with the material, identify gaps in understanding, and get used to the style of mathematical reasoning that will be useful in the PhD. They are rigorous, but they are not intended to be punitive or excessively difficult. You should think of them as a structured way to learn the material rather than as a hurdle designed to discourage you.

The TA will provide feedback on submitted problem sets, but the problem sets themselves are not graded. Solutions will also be posted early. Even so, working through the assignments is strongly encouraged, both for practice and for building a study group with your classmates, which is crucial for your success in the first year and beyond.

Course Topics

Topics marked with a dag (†) are skimmed and should be reviewed (either from textbooks or the video lectures) before the class starts. Topics with a star (*) may not be covered within the schedule and, if omitted, will not be tested on the exam.

If you do not yet feel comfortable with the subjects listed below, regular attendance and consistent work are strongly recommended.

Linear Algebra

- Matrix algebra (†)
- Solutions to general linear systems (†)
- Vectors and vector spaces (†)
- Linear maps and their matrix representations
- Eigenvectors, eigenvalues and diagonalization
- Diagonalizability and the Jordan canonical form (*)
- Inner product spaces and orthogonality
- Spectral decomposition of symmetric (unitary) matrices
- Quadratic forms and positive definiteness
- Further matrix decompositions (*)

Analysis in Euclidean Spaces

- Sets, relations, and functions (†)

- Countability and cardinality (*)
- Open and closed sets
- Sequences and convergence (†)
- Cauchy sequences, completeness and contraction mappings
- Continuity of functions (†)
- Compactness and uniform continuity
- Sequences of functions and uniform convergence

Multivariable Calculus

- Derivatives (†) and differentiability of functions
- Mean-value theorems
- Implicit function theorem and inverse function theorem
- Taylor expansions and log-linearization.
- Homogeneous function and its characterization
- Riemann integration (†)

Convexity

- Convex sets (†)
- Separation theorems for convex sets (*)
- Convex and concave functions
- Quasi-convex and quasi-concave functions

Optimization and Correspondences

- Unconstrained optimization under differentiability (†)
- Constrained optimization: Kuhn-Tucker Theorem
- Lagrange duality (*)
- Envelope theorem and comparative statics under differentiability
- Upper- and lower-hemicontinuity of correspondences
- Berge's maximum theorem
- Fixed point theorems of Brouwer and Kakutani

Probability Theory

- Probability and random variables (†)
- Expectation and conditional expectation
- Law of large numbers and central limit theorem

I am looking forward to being your instructor this year. If you have any questions about anything at all, please do not hesitate to reach out.

Schedule Overview

Sun	Mon	Tue	Wed	Thu	Fri	Sat
08/09	08/10 Lecture: 9am–12pm	08/11 Lecture: 9am–12pm	08/12 Lecture: 9am–12pm Recitation: 2–3:30pm	08/13 Lecture: 9am–12pm Joe's OH: 2–3pm	08/14 Lecture: 9am–12pm Hao's OH: 2–3pm Recitation: 3–4:30pm	08/15
08/16	08/17 Lecture: 9am–12pm **Assignment 1 Due**	08/18 Lecture: 9am–12pm	08/19 Lecture: 9am–12pm Recitation: 2–3:30pm	08/20 Lecture: 9am–12pm Joe's OH: 2–3pm	08/21 Lecture: 9am–12pm Hao's OH: 2–3pm Recitation: 3–4:30pm	08/22
08/23	08/24 Lecture: 9am–12pm	08/25 Lecture: 9am–12pm Joe's OH: 2–3pm	08/26 Lecture: 9am–12pm Recitation: 2–4:00pm **Assignment 2 Due**	08/27 Lecture: 9am–12pm Hao's OH: 2–4pm	08/28 **Exam: 9am–12pm (In-Person Closed-Book Exam)**	08/29